

DP-588

Specifications



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1. Overview

■ safety instruction

Read the following notes carefully before using the printer.

■ warn

 The print head is a heating part. During the printing process and the printing, do not touch the print head and surrounding parts.

 Do not touch the print head and the connection plug-in to avoid damage to the print head due to static electricity.

■ matters need attention

- 1) The printer shall be installed in a flat and stable place.
- 2) Avoid placing the printer in a place of vibration and impact.
- 3) Leave enough space around the printer for maintenance and operation.
- 4) The printer should be far away from the water source and avoid direct sunlight, strong light and heat source.
- 5) Do not use and save printers in high temperature, high humidity, and highly polluted places.
- 6) If the printer will not be used for an extended period of time, disconnect the printer power.
- 7) The printer shall not be printed without paper, otherwise the print head will be seriously damaged.
- 8) In order to ensure printing quality and product life, recommended or equivalent quality paper is recommended.
- 9) If the use requirements are met, it is recommended to set the low-level of printing concentration as far as possible to avoid affecting the use of the printing head life span.
- 10) Moist air is not allowed to condense on the surface of the printer, if it has formed, do not turn on the power of the printer until the dew has been removed.
- 11) Avoid water or conductive substances (e. g., metal) from entering the printer and turn off the power immediately once it occurs.
- 12) When plugging in or disconnecting each interface, the power supply must be turned off, otherwise it may cause damage to the printer control circuit.
- 13) Users shall not disassemble the printer for maintenance without authorization.
- 14) Keep this manual properly for use and reference.

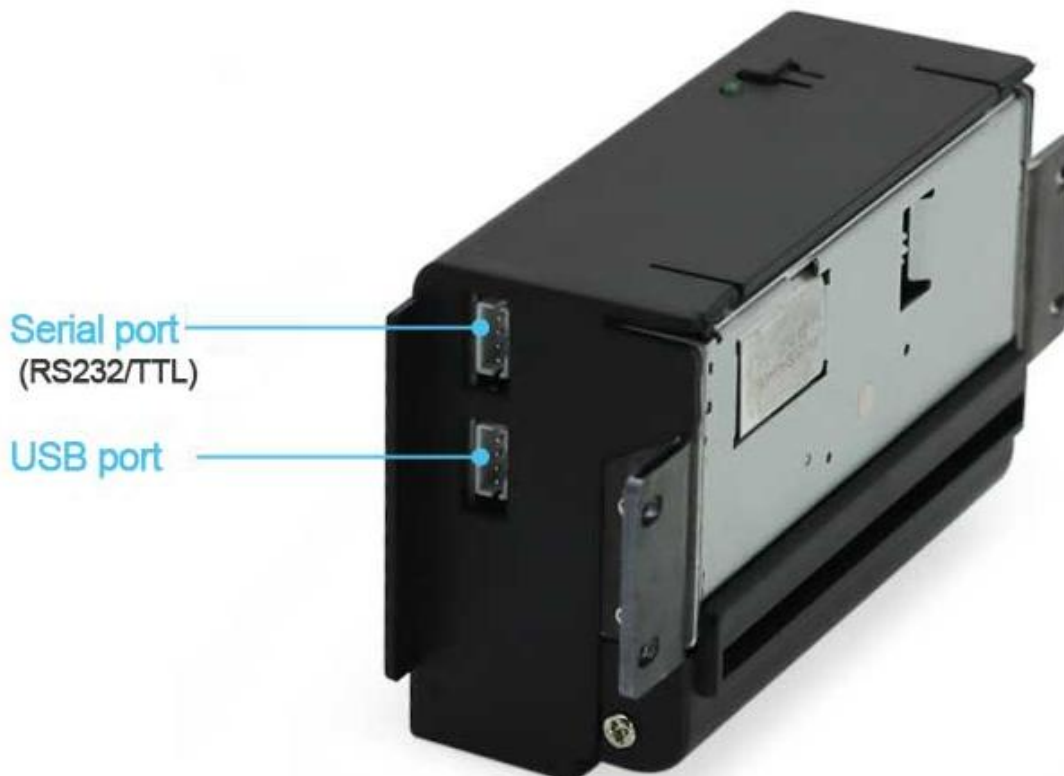
DP-588 can support thermal label paper and thermal continuous paper, the maximum paper width is 79mm. This product is mainly used for receipts from the host computer, and can also be used or receipts need to be printed. 588A connects devices through TTL/RS232/USB interface.

1.1 Main features

- With cutter function, you can set full cut or half cut.
- Integrated mechanism design
- Plug and play, easy to use
- Low noise and reliable performance
- Outer diameter of paper roll ϕ 79mm
- Out of paper status will automatically reply, the honey device will beep three times and the LED light will be three times when the paper is out, and the Out of paper status data will be returned to the host computer at the same time.
- The status of printing completion and printing failure will be automatically replied, and the printing completion status data will be returned to the host computer when printing is completed.

- The maximum printing speed is 60mm/s

2. Main technical specifications



2.1 Technical specifications



Product number	DP-588
Communication Interface	TTL/RS232+USB
Baud rate	Default 115200
Operating Voltage	12V
Working current	Normal working current 1.8-2.2A peak current 2A
printing method	thermal printing
print color	black and white output
Resolution	203dpi 8 dots/mm 576 dots per line
printing speed	60mm/s
roll type	Thermal paper/Label paper
Roll Specifications	80mm $\pm$ 0.05mm (width) 0.05~0.1mm (thickness)
Paper roll diameter	Paperless warehouse design
paper cutting method	Automatic paper cutting support full cut half cut
service life	50 km
Product color	black
Platform support	Windows, linux, and Android, STM32, 51 MCU
print content	Text, graphics, characters, barcodes, QR codes

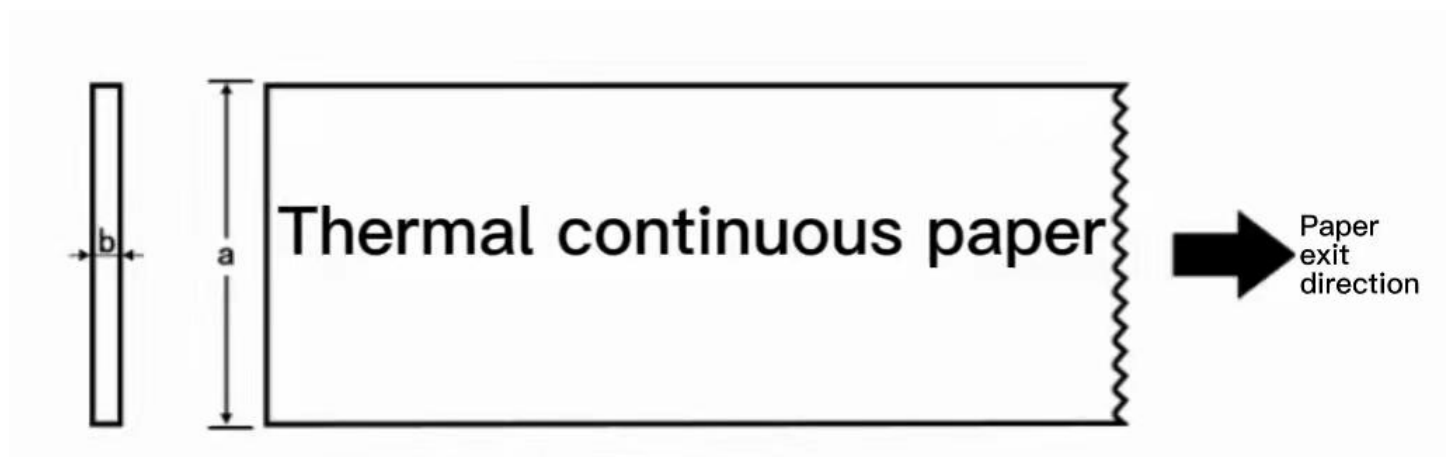
Remark	The USB interface supports the computer system such as XP/Win7/Win8, which can use the driver to print, and the USB/serial communication can be connected to the Android tablet for printing. It provides the Android development kit and provides technical support guidance.
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2.2 Printing Consumables Specifications

2.2.1 Paper roll parameters

- Paper type: thermal label paper, thermal continuous paper
- Thermal layer: external
- Paper width: 40mm-80mm

2.2.2 Continuous paper parameters



Code	meaning	Maximum (mm)	Minimum (mm)
a	paper width	79	/
b	paper roll thickness	0.1	0.05

Note: Do not scratch the print head and the rubber roller with hard objects (such as tweezers) when cleaning the print head or the paper feed roller; in order to improve the service life of the print head, print on self-adhesive printing paper, and do not dry it for regular printing Adhesive paper will have residual glue on the back of the adhesive. It is recommended to clean the print head and rubber roller every month. If the environment is harsh, increase the cleaning appropriately.

The print roller should be cleaned when the print head experiences any of the following:

- The printing is not clear;
- Paper feed noise is loud.

The cleaning steps of the printing rubber roller are as follows

- Turn off the printer, slide the upper cover limit block, and open the printer upper cover assembly gear;
- Rotate the print roller, and at the same time use an alcohol cotton ball (should be wrung out) to wipe off the dust and

stains on the surface of the print roller;

Wait 5-10 minutes for the alcohol to evaporate completely, then close the printer.

3. printer loading paper



The paper loading method must be the same as the picture above, so that the content can be printed correctly, otherwise the content cannot be printed, and only blank paper will be taken.

4. Printer Interface Definition



4.1 2pin interface input power supply 12V, the current is above 2.5A.

4.2 The serial port of the printer has two types: TTL and 232. The default baud rate is 115200. The following is the serial port wiring method.

Printer serial port	Host computer device serial port
GND	GND
RX	TX
TX	RX
DTR	empty (no answer)

4.3 When using the USB port to communicate with the computer, the USB port of the printer is a virtual serial port, and the serial port debugging assistant can be used to send data to the printer. If the printer serial port is not recognized on the computer. That is, the printer driver is not installed automatically, you need to install the driver manually, please refer to the driver installation manual.

5. Print a self-test page

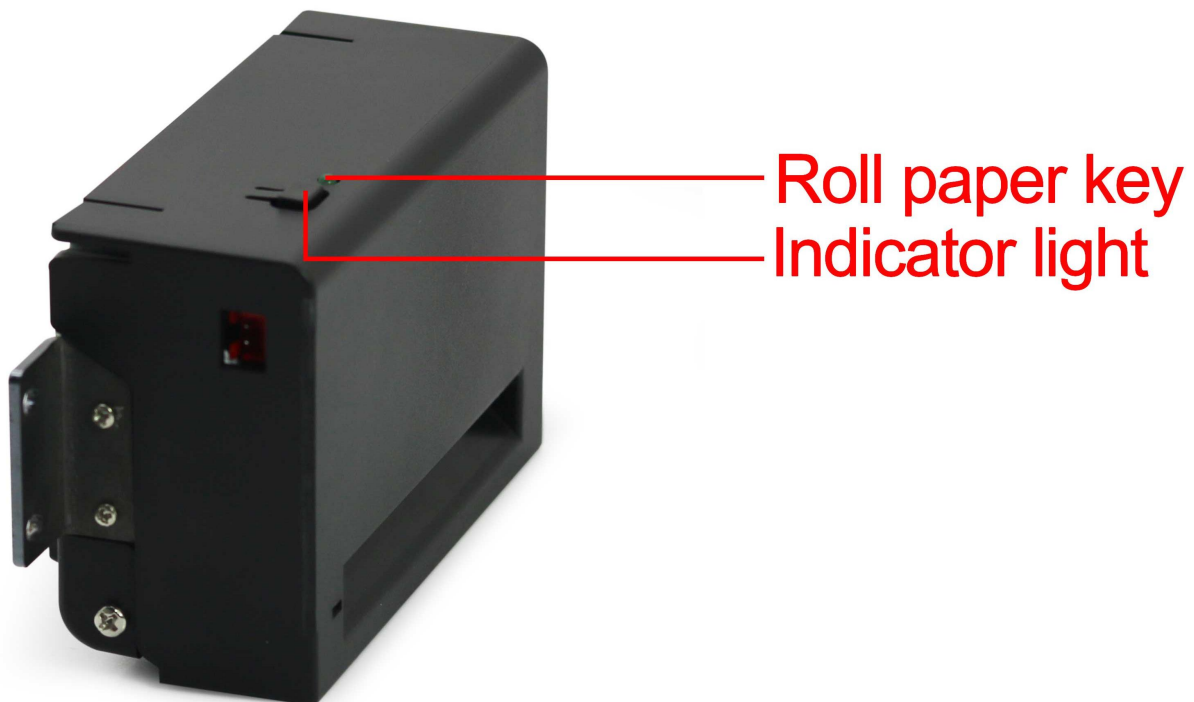
5.1 Press and hold the button for more than 3s to automatically print out a test page, and there are related printer parameters in the test page. Pressing the button will empty out a blank piece of paper.

5.2 Switch between labels and receipts by sending instructions to the printer to switch

Send hexadecimal data to set label mode: 1F 2D 4D 01 01

Send data to set receipt mode: 1F 2D 4D 01 02

5.3 The green light is always on, normal, the green light of Out of paper status will flash three times and stop.



6. Detailed explanation of receipt instructions

6.1 Initialize the printer

Command name	Initialize the printer
Instruction code	ASCII : ESC @ Decimal : 27 64 Hex : 1B 40
Function description	Initialize the printer the following: clear print cache Each parameter restores Defaults
Parameter range	None
Defaults	None
Support model	All model
Precautions	None
Example of use	None

6.2 Print a self-test page

Command name	Print the self-test page
Instruction code	ASCII : DC2 T Decimal : 18 94 Hex : 12 54
Function description	The printer prints a self-test page, which contains the program version of the printer, the type of communication interface, the code page and some other data
Parameter range	None
Defaults	None
Support model	All model
Precautions	None
Example of use	1B 40 12 54

6.3Set character printing method

Command name	Set character printing method																														
Instruction code	ASCII : ESC ! n Decimal : 27 33 n Hex : 1B 21 n																														
Function description	Set the character printing mode (type, antiwhite, inversion, bold, height, width, and underscore) bits of parameter n Definition is as follows: <table><tr><th>Bit</th><th>function</th><th>value</th></tr><tr><td></td><td></td><td>0 1</td></tr><tr><td>0</td><td>Type</td><td>Normal small characters</td></tr><tr><td>1</td><td>Undefined</td><td></td></tr><tr><td>2</td><td>Undefined</td><td></td></tr><tr><td>3</td><td>Bold</td><td>cancel the setting</td></tr><tr><td>4</td><td>x High</td><td>cancel the setting</td></tr><tr><td>5</td><td>x width</td><td>cancel the setting</td></tr><tr><td>6</td><td>Undefined</td><td></td></tr><tr><td>7</td><td>Underline</td><td>cancel the setting</td></tr></table>	Bit	function	value			0 1	0	Type	Normal small characters	1	Undefined		2	Undefined		3	Bold	cancel the setting	4	x High	cancel the setting	5	x width	cancel the setting	6	Undefined		7	Underline	cancel the setting
Bit	function	value																													
		0 1																													
0	Type	Normal small characters																													
1	Undefined																														
2	Undefined																														
3	Bold	cancel the setting																													
4	x High	cancel the setting																													
5	x width	cancel the setting																													
6	Undefined																														
7	Underline	cancel the setting																													
Parameter range	None																														
Defaults	n = 0																														
Support model	All model																														
Precautions	This instruction is valid for both Chinese and foreign language fonts When ESC @, printer reset, power off, the setting of this instruction fails																														
Example of use	1B 40 1B 21 01 30 31 32 0D 0A 1B 40 1B 21 02 30 31 32 0D 0A 1B 40 1B 21 04 30 31 32 0D 0A 1B 40 1B 21 08 30 31 32 0D 0A 1B 40 1B 21 10 30 31 32 0D 0A 1B 40 1B 21 20 30 31 32 0D 0A 1B 40 1B 21 40 30 31 32 0D 0A 1B 40 1B 21 80 30 31 32 0D 0A																														

6.4set character size

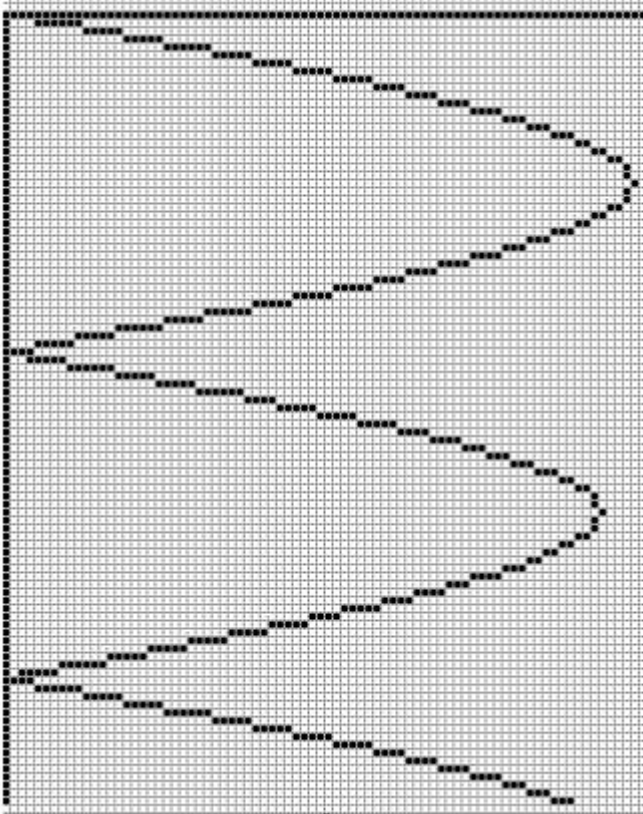
Command name	Sets the character size
Instruction code	ASCII : GS ! n Decimal : 29 33 n Hex : 1d 21 n

Function description	1d 21 00 normal font (default) 1d 21 11 x height x width 1d 21 10 font X width 1d 21 01 font X height
Parameter range	None
Defaults	n = 0
Support model	All model
Precautions	This instruction is valid for both Chinese and foreign fonts other than HRI characters When ESC @, printer reset, power off, the setting of this instruction fails
Example of use	1b 40 1d 21 00 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 0d 0a 1b 40 1d 21 11 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 0d 0a 1b 40 1d 21 10 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 0d 0a 1b 40 1d 21 01 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 0d 0a

6.5Set print alignment

Command name	Sets the print alignment(left, center, right)
Instruction code	ASCII : ESC a n Decimal : 27 97 n Hex : 1B 61 n
Function description	All data in a row is aligned, and the meaning of the n value is as follows: n pattern 0,48 To the left 1,49 Middle House 2,50 To the right
Parameter range	$0 \leq n \leq 2$ Or $48 \leq n \leq 50$
Defaults	n = 0
Support model	All model
Precautions	When ESC @, printer reset, power off, the setting of this instruction fails
Example of use	1B 40 1B 61 00 C4 AC C8 CF D7 F3 B6 D4 C6 EB 0D 0A 1B 40 1B 61 01 BE D3 D6 D0 B6 D4 C6 EB 0D 0A 1B 40 1B 61 02 BF BF D3 D2 B6 D4 C6 EB 0D 0A

6.6Print line segments in horizontal position (curve print command)

Command name	Print line segments horizontally (Curve Print command)
Instruction code	ASCII : GS ' n x1sL x1eH x1eL x1eH ...xnsL xnsH xneL xneH Decimal : 1D 27 n x1sL x1eH x1eL x1eH ...xnsL xnsH xneL xneH Hex : 29 39 n x1sL x1eH x1eL x1eH ...xnsL xnsH xneL xneH
Function description	An enlarged print is shown below: Each horizontal curve segment can be viewed as consisting of these points with segment length 1. To print n lines of horizontal line segments, use this command continuously to print out the desired curve.  xksL : The low-level horizontal coordinate of the starting point of the K-line; xksH : The high-order horizontal coordinate of the starting point of the K-line; xkeL : The low-level horizontal coordinate of the end point of the K-line; xkeH : The high-order horizontal coordinate of the end point of the K-line; The coordinate start position is usually to the left of the print area.The minimum coordinate coordinate is (0,0), the maximum horizontal coordinate value is 383, xkeL + xkeH * 256 Line data can not be arranged in the order within the specified range; <pre>Char SendStr[8]; Char SendStr2[16]; Float i; Short y1,y2,y1s,y2s; // Print the Y-axis (one line) SendStr[0]=0x1D;</pre>

```

SendStr[1]=0x27;
SendStr[2] =1; // one row
SendStr[3]=30
SendStr[4] =0; // Start point
SendStr[5]=104;
SendStr[6] =1; // End point
PreSendData(SendStr,7);

//Print curve
SendStr[0]=0x1D;
SendStr[1]=0x27;
SendStr[2] =3; // Three lines: X-axis, sin and cos function curve
three lines: X-axis, sin, and cos
function
SendStr[3] =180; SendStr[4] =0; // X axis position
SendStr[5]=180; SendStr[6]=0;
for(i=1;i<1200;i++)
{
For y1=sin (i / 180 * 3.1416) * (380-30) / 2 + 180; // calculate the sin
function coordinates
For y2=cos (i / 180 * 3.1416) * (380-30) / 2 + 180; // calculate the cos
function coordinates
If(i==1){y1s=y1;y2s=y2;}
PreSendData(SendStr,7);

If(y1s<y1)
{
PreSendData (& y1s, 2); // sin function is at the starting point of the line
The PreSendData (& y1,2); // sin function is at the end point of the line
}
Else
{
The PreSendData (& y1,2); // sin function is at the starting point of the line
The PreSendData (& y1s, 2); // sin function is at the end point of the
line
}
If(y2s<y2)
{
PreSendData (& y2s, 2); // cos function is at the starting point of the line
The PreSendData (& y2,2); // cos function is at the end point of the line
}
Else
{
PreSendData (& y2,2); // cos function is at the starting point of the line
The PreSendData (& y2s, 2); // cos function is at the end point of the
line
}
The y1s=y1; // When the print enters the next row, the sin function

```

	curve starts in the abscissa The y2s=y2; // When the print goes to the next row, the cos function curve starts in the abscissa }
Parameter range	$0 \leq n \leq 8$
Defaults	None
Support model	Portable printer
Precautions	When printing a point, xkeL=xksL, xkeH=xksH
Example of use	<pre> 1d 27 01 00 00 00 00 1d 27 01 01 00 0f 00 1d 27 01 10 00 1f 00 1d 27 01 20 00 2c 00 1d 27 01 2d 00 3a 00 1d 27 01 3b 00 44 00 1d 27 01 45 00 4c 00 1d 27 01 4d 00 54 00 1d 27 01 55 00 5c 00 1d 27 01 5d 00 63 00 1d 27 01 64 00 6a 00 1d 27 01 6b 00 71 00 1d 27 01 72 00 77 00 1d 27 01 78 00 7d 00 1d 27 01 7e 00 84 00 1d 27 01 85 00 8a 00 1d 27 01 8b 00 91 00 1d 27 01 92 00 97 00 1d 27 01 98 00 9d 00 1d 27 01 9e 00 a3 00 1d 27 01 a4 00 a9 00 1d 27 01 aa 00 af 00 1d 27 01 b0 00 b4 00 1d 27 01 b5 00 b9 00 1d 27 01 ba 00 bf 00 1d 27 01 c0 00 c4 00 1d 27 01 c5 00 c9 00 1d 27 01 ca 00 cf 00 1d 27 01 d0 00 d4 00 1d 27 01 d5 00 d8 00 1d 27 01 d9 00 dc 00 1d 27 01 dd 00 df 00 1d 27 01 e0 00 e3 00 1d 27 01 e4 00 e6 00 1d 27 01 e7 00 e9 00 1d 27 01 ea 00 ec 00 1d 27 01 ed 00 ef 00 1d 27 01 f0 00 f1 00 1d 27 01 f2 00 f3 00 1d 27 01 f4 00 f5 00 1d 27 01 f6 00 f7 00 1d 27 01 f8 00 f8 00 1d 27 01 f9 00 fa 00 1d 27 01 fb 00 fb 00 1d 27 01 fc 00 fd 00 1d 27 01 fe 00 fe 00 1d 27 01 ff 00 ff 00 1d 27 01 00 01 00 01 1d 27 01 01 01 01 01 1d 27 01 02 01 02 01 1d 27 01 03 01 03 01 1d 27 01 04 01 04 01 1d 27 01 05 01 05 01 1d 27 01 06 01 06 01 1d 27 01 06 01 06 01 1d 27 01 07 01 07 01 1d 27 01 07 01 07 01 1d 27 01 07 01 07 01 1d 27 01 07 01 07 01 1d 27 01 07 01 07 01 1d 27 01 06 01 06 01 1d 27 01 06 01 06 01 1d 27 01 05 01 05 01 1d 27 01 04 01 04 01 1d 27 01 04 01 04 01 1d 27 01 03 01 03 01 1d 27 01 02 01 02 01 1d 27 01 00 01 00 01 1d 27 01 ff 00 ff 00 1d 27 01 fe 00 fe 00 1d 27 01 fc 00 fd 00 1d 27 01 f9 00 fa 00 1d 27 01 f8 00 f8 00 </pre>

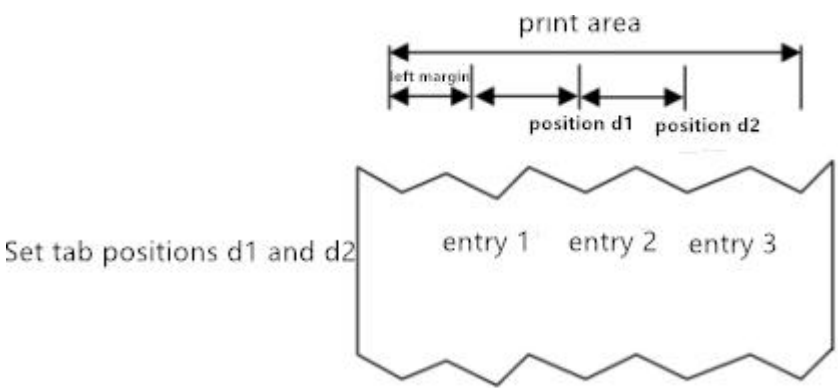
1d 27 01 f6 00 f7 00 1d 27 01 f4 00 f5 00
1d 27 01 f2 00 f3 00 1d 27 01 f0 00 f1 00
1d 27 01 ed 00 ef 00 1d 27 01 ea 00 ec 00
1d 27 01 e7 00 e9 00 1d 27 01 e4 00 e6 00
1d 27 01 e0 00 e3 00 1d 27 01 dd 00 df 00
1d 27 01 d9 00 dc 00 1d 27 01 d5 00 d8 00
1d 27 01 d0 00 d4 00 1d 27 01 ca 00 cf 00
1d 27 01 c5 00 c9 00 1d 27 01 c0 00 c4 00
1d 27 01 ba 00 bf 00 1d 27 01 b5 00 b9 00
1d 27 01 b0 00 b4 00 1d 27 01 aa 00 af 00
1d 27 01 a4 00 a9 00 1d 27 01 9e 00 a3 00
1d 27 01 98 00 9d 00 1d 27 01 92 00 97 00
1d 27 01 8b 00 91 00 1d 27 01 85 00 8a 00
1d 27 01 7e 00 84 00 1d 27 01 78 00 7d 00
1d 27 01 72 00 77 00 1d 27 01 6b 00 71 00
1d 27 01 64 00 6a 00 1d 27 01 5d 00 63 00
1d 27 01 55 00 5c 00 1d 27 01 4d 00 54 00
1d 27 01 45 00 4c 00 1d 27 01 3b 00 44 00
1d 27 01 2d 00 3a 00 1d 27 01 20 00 2c 00
1d 27 01 10 00 1f 00 1d 27 01 01 00 0f 00
1d 27 01 00 00 00 00 1d 27 01 00 00 00 00
1d 27 01 01 00 0f 00 1d 27 01 10 00 1f 00
1d 27 01 20 00 2c 00 1d 27 01 2d 00 3a 00
1d 27 01 3b 00 44 00 1d 27 01 45 00 4c 00
1d 27 01 4d 00 54 00 1d 27 01 55 00 5c 00
1d 27 01 5d 00 63 00 1d 27 01 64 00 6a 00
1d 27 01 6b 00 71 00 1d 27 01 72 00 77 00
1d 27 01 78 00 7d 00 1d 27 01 7e 00 84 00
1d 27 01 85 00 8a 00 1d 27 01 8b 00 91 00
1d 27 01 92 00 97 00 1d 27 01 98 00 9d 00
1d 27 01 9e 00 a3 00 1d 27 01 a4 00 a9 00
1d 27 01 aa 00 af 00 1d 27 01 b0 00 b4 00
1d 27 01 b5 00 b9 00 1d 27 01 ba 00 bf 00
1d 27 01 c0 00 c4 00 1d 27 01 c5 00 c9 00
1d 27 01 ca 00 cf 00 1d 27 01 d0 00 d4 00
1d 27 01 d5 00 d8 00 1d 27 01 d9 00 dc 00
1d 27 01 dd 00 df 00 1d 27 01 e0 00 e3 00
1d 27 01 e4 00 e6 00 1d 27 01 e7 00 e9 00
1d 27 01 ea 00 ec 00 1d 27 01 ed 00 ef 00
1d 27 01 f0 00 f1 00 1d 27 01 f2 00 f3 00
1d 27 01 f4 00 f5 00 1d 27 01 f6 00 f7 00
1d 27 01 f8 00 f8 00 1d 27 01 f9 00 fa 00
1d 27 01 fb 00 fb 00 1d 27 01 fc 00 fd 00
1d 27 01 fe 00 fe 00 1d 27 01 ff 00 ff 00
1d 27 01 00 01 00 01 1d 27 01 01 01 01 01
1d 27 01 02 01 02 01 1d 27 01 03 01 03 01
1d 27 01 04 01 04 01 1d 27 01 05 01 05 01
1d 27 01 06 01 06 01 1d 27 01 06 01 06 01

	1d 27 01 07 01 07 01 1d 27 01 07 01 07 01
	1d 27 01 07 01 07 01 1d 27 01 07 01 07 01
	1d 27 01 07 01 07 01 1d 27 01 06 01 06 01
	1d 27 01 06 01 06 01 1d 27 01 05 01 05 01
	1d 27 01 04 01 04 01 1d 27 01 04 01 04 01
	1d 27 01 03 01 03 01 1d 27 01 02 01 02 01
	1d 27 01 00 01 00 01 1d 27 01 ff 00 ff 00
	1d 27 01 fe 00 fe 00 1d 27 01 fc 00 fd 00
	1d 27 01 f9 00 fa 00 1d 27 01 f8 00 f8 00
	1d 27 01 f6 00 f7 00 1d 27 01 f4 00 f5 00
	1d 27 01 f2 00 f3 00 1d 27 01 f0 00 f1 00
	1d 27 01 ed 00 ef 00 1d 27 01 ea 00 ec 00
	1d 27 01 e7 00 e9 00 1d 27 01 e4 00 e6 00
	1d 27 01 e0 00 e3 00 1d 27 01 dd 00 df 00
	1d 27 01 d9 00 dc 00 1d 27 01 d5 00 d8 00
	1d 27 01 d0 00 d4 00 1d 27 01 ca 00 cf 00
	1d 27 01 c5 00 c9 00 1d 27 01 c0 00 c4 00
	1d 27 01 ba 00 bf 00 1d 27 01 b5 00 b9 00
	1d 27 01 b0 00 b4 00 1d 27 01 aa 00 af 00
	1d 27 01 a4 00 a9 00 1d 27 01 9e 00 a3 00
	1d 27 01 98 00 9d 00 1d 27 01 92 00 97 00
	1d 27 01 8b 00 91 00 1d 27 01 85 00 8a 00
	1d 27 01 7e 00 84 00 1d 27 01 78 00 7d 00
	1d 27 01 72 00 77 00 1d 27 01 6b 00 71 00
	1d 27 01 64 00 6a 00 1d 27 01 5d 00 63 00
	1d 27 01 55 00 5c 00 1d 27 01 4d 00 54 00
	1d 27 01 45 00 4c 00 1d 27 01 3b 00 44 00
	1d 27 01 2d 00 3a 00 1d 27 01 20 00 2c 00
	1d 27 01 10 00 1f 00 1d 27 01 01 00 0f 00
	1d 27 01 00 00 00 00

6.7Set the horizontal tab position

Command name	horizontal tabulation
Instruction code	ASCII : HT Decimal : 9 Hex : 09
Function description	Move the print location to the next tabulation location
Parameter range	None
Defaults	None
Support model	All model
Precautions	The tab position is set by ESC D If the tab position is not set (default no horizontal tabulation

	position), this instruction will be regarded as an LF instruction If the table position exceeds the print area, the coordinates are moved to the starting position of the next line (as full, print and line feed)
--	---

Command name	
Instruction code	ASCII : ESC D [d]k NUL Decimal : 27 68 [d]k 0 Hex : 1B 44 [d]k 00
Function description	Set the horizontal tabulation position with the following parameters: d1 ...The dk: Horizontal table position ,per 8 points as one unit and NULL as the end character
Parameter range	XX58: $1 \leq d \leq 46$ ($d_1 < d_2 < \dots < d_k$, $1 \leq k \leq 16$) XX80: $1 \leq d \leq 70$ ($d_1 < d_2 < \dots < d_k$, $1 \leq k \leq 16$)
Defaults	[d] k = 0 (Default no level table location)
Support model	All model
Precautions	The tab positions are indicated as follows:  Set tab positions d1 and d2 Up to 16 tabulation locations are supported Using this instruction will cancel previous settings for TAB locations Use k for purposes, without transmission Transfer[d]k is considered finished when it encounters NULL If dk is less than or equal to dk-1, it is considered ended, and the remaining data is regarded as normal data processing The tabtable position can be switched by HT When the left distance is changed, the balance sheet position changes simultaneously When ESC @, printer reset, power off, the setting of this instruction fails
Example of use	1B 44 0B 12 19 00 0D 0A 20 20 20 C6 B7 20 C3 FB 09 B5 A5 BC DB 09 CA FD C1 BF 09 BD F0 B6 EE 09 0D 0A C5 A3 C8 E2 CB C9 D0 A1 B1 B4 0D 0A 09 31 2E 30 09 32 09 32 2E 30 30 0D 0A C1 F1 C1 AB B5 B0 CC A2 0D 0A 09 31 30 32 2E 30 09 32 09 32 30 34 2E 30 30 0D 0A D7 CF CA ED D4 B2 D4 B2 CB D8 0D 0A 09 39 31 2E 30 09 32 30 09 31 38 32 30 2E 30 30 0D 0A Instructions: 1B 44 0B 12 19 00 ==0B is the width of the first column, 12 is the second column

width, 19 is the third column width, and the remaining width is the last column width ending with a 00 split

First column 0B	Second case 12	Third column 19	last column

is like the width of a row

0 D, 0 A, change the row

20 20 20 C6 B7 20 C3 FB ==02 is a space, the first column of text content "product name"

09 ==09 As the column symbol,

B5 A5 BC DB The second column of text content "unit price"

09 The column symbol

CA FD C1 BF The third case of the text content "quantity"

09 The column symbol

BD F0 B6 EE fourth column text content "Amount"

09 The column symbol

0 D, 0 A, change the row

C5 A3 C8 E2 CB C9 D0 A1 B1 B4 first column text content "Beef Pine Baby"

0 D, 0 A, change the row

09 The column symbol31 2E 30 Second case of 1.0

09 The column symbol32 The third column is the 2

09 The column symbol32 2E 30 30 4th column 2.00

0 D, 0 A, change the row

C1 F1 C1 AB B5 B0 CC A2 first column "Durian Egg Tart"

0D 0A

09 The column symbol31 30 32 2E 30 Second Case 102.0

09 The column symbol32 The third column is the 2

09 The column symbol32 30 34 2E 30 30 Fourth Column 204.00

0 D, 0 A, change the row

D7 CF CA ED D4 B2 D4 B2 CB D8 first column "purple potato Yuanyuan"

0 D, 0 A,change the row

09 The column symbol39 31 2E 30 Second Case 91.0

09 The column symbol3,230 The third column is the number 20

09 The column symbol31 38 32 30 2E 30 30 Fourth Column 1820.00

0 D, 0 A,change the row

6.81D barcode printing instructions

6.8.1Set the 1D barcode readable characters (HRI) print position

Command name	Set the barcode-readable character (HRI) print location
Instruction code	ASCII : GS H n Decimal: 2,972 n Hex: 1D 48 n
Function description	Set the barcode readable character (HRI) print position, the n parameter meaning is as follows: n PP(print position) 0,48 No printing occurs 1,49 Above the bar code 2,50 Below the bar code 3,51 Above and below the barcode
Parameter range	$0 \leq n \leq 3$ OR $48 \leq n \leq 51$
Defaults	n = 0
Support model	All models
Precautions	When ESC @, printer reset, power off, the setting of this instruction fails
Example of use	None

6.8.2Set 1D barcode height

Command name	Set the 1-dimensional barcode height
Instruction code	ASCII : GS h n Decimal: 29,104 n Hex: 1D 68 n
Function description	Set the height of the bar code to n point, and the parameter n means as follows:  Height: 50  Height: 100
Parameter range	$1 \leq n \leq 255$
Defaults	n = 64
Support model	All models
Precautions	When ESC @, printer reset, power off, the setting of this instruction fails

Example of use	None
----------------	------

6.8.3 Set the 1D barcode width

Command name	Set the one-dimensional barcode width
Instruction code	ASCII : GS w n Decimal: 29,119 n Hex: 1D 77 n
Function description	Set the barcode unit to n point, and the parameter n means as follows: 
Parameter range	$1 \leq n \leq 6$
Defaults	n = 2
Support model	All models
Precautions	When ESC @, printer reset, power off, the setting of this instruction fails
Example of use	None

6.8.4 Printing 1D barcodes

Command name	
Instruction code	(A) ASCII : GS k m [d]k NUL Decimal: 29,107 m [d] k NUL Hex: 1D 6B m [d] k NUL (B) ASCII : GS k m n [d]k Decimal: 29,107 m n [d] k Hex: 1D 6B m n [d] k
Function description	Print the one-dimensional barcode, the significance of each parameter is as follows: The m is the encoding mode The n is the length of the coded data, used only in (B) mode, (A) and (B) instructions where the data segment of (A) ends with the NULL character and (B) indicates the length of the data The [d] k is the barcode data

The k is the length of the barcode data and is used for indication and without transmission

The relationship between the parameters is shown in the following table:

(instruct A)

m	coding system	Barcode data (SP for spaces)			
		DL	k	character set	data (d)
0	UPC-A	fixed	k = 11, 12	0~9	$48 \leq d \leq 57$
1	UPC-E	fixed	$6 \leq k \leq 8$, k = 11, 12	0~9	$48 \leq d \leq 57$ [When k = 7,8,11,12, d1 = 48]
2	JAN13 (EAN13)	fixed	k = 12, 13	0~9	$48 \leq d \leq 57$
3	JAN8 (EAN8)	fixed	k = 7, 8	0~9	$48 \leq d \leq 57$
4	CODE39	variable	$1 \leq k \leq 255$	0~9, A~Z SP, \$, %, +, -, ., /	$48 \leq d \leq 57$, $65 \leq d \leq 90$, d = 32, 36, 37, 42, 43, 45, 46, 47
5	ITF (Interleaved 2 of 5)	variable	$2 \leq k \leq 255$ (even number)	0~9	$48 \leq d \leq 57$
6	CODABAR (NW-7)	variable	$1 \leq k$	0~9, A~D, a~d \$, +, -, ., /, :	$48 \leq d \leq 57$, $65 \leq d \leq 68$, $97 \leq d \leq 100$, d = 36, 43, 45, 46, 47, 58 ($65 \leq d1 \leq 68$, $65 \leq dk \leq 68$, $97 \leq d1 \leq 100$, $97 \leq dk \leq 100$)

(instruct B)

m	coding system	Barcode data (SP for spaces)			
		DL	n	character set	data (d)
65	UPC-A	fixed	n = 11, 12	0~9	$48 \leq d \leq 57$
66	UPC-E	fixed	$6 \leq n \leq 8$, n = 11, 12	0~9	$48 \leq d \leq 57$ [When n = 7,8,11,12, d1 = 48]
67	JAN13 (EAN13)	fixed	n = 12, 13	0~9	$48 \leq d \leq 57$
68	JAN8 (EAN8)	fixed	n = 7, 8	0~9	$48 \leq d \leq 57$
69	CODE39	variable	$1 \leq n \leq 255$	0~9, A~Z SP, \$, %, +, -, ., :	$48 \leq d \leq 57$, $65 \leq d \leq 90$,

					/	d = 32, 36, 37, 42, 43, 45, 46, 47
	70	ITF (Interleaved 2 of 5)	variable	$1 \leq n \leq 255$ (even number)	0~9	$48 \leq d \leq 57$
	71	CODABAR (NW-7)	variable	$1 \leq n \leq 255$	0~9, A~D, a~d, \$, +, -, ., /, :	$48 \leq d \leq 57$, $65 \leq d \leq 68$, $97 \leq d \leq 100$, d = 36, 43, 45, 46, 47, 58 ($65 \leq d1 \leq 68$, $65 \leq dk \leq 68$, $97 \leq d1 \leq 100$, $97 \leq dk \leq 100$)
	72	CODE93	variable	$1 \leq n \leq 255$	00H~7FH	$0 \leq d \leq 127$
	73	CODE128	variable	$2 \leq n \leq 255$	00H~7FH	$0 \leq d \leq 127$
	74	UCC/EAN 128	variable	$2 \leq n \leq 255$	00H~7FH C1H~C4H(FNC)	$0 \leq d \leq 127$ d = 193, 194, 195, 196
Parameter range	(A) $0 \leq m \leq 6$ (B) $65 \leq m \leq 74$					
Defaults	None					
Support model	All model					
Precautions	If the barcode width exceeds the printable area, the printer does not perform barcode printing When this instruction is executed as required, not affected by ESC 2, ESC 3 line spacing setting This instruction is not subject to the ESC!Character style setting influence After this instruction is executed, the print location is restored to the print start position The m parameters 0 ~ 6 (A) and 65 ~ 71 (B) select the same coding system with the same printing effect For the m parameters 0 ~ 6 (A), the barcode data ends in NULL For the m parameters 65 ~ 74 (B), the barcode data in n indicates the data length Use k for signals, and no transmission is required When printing a UPCA (m = 0 or 65), note that: Whether the input data length is 11 or 12, the check bit is automatically inserted or error corrected Start character, middle separator, and end character is automatically inserted When printing a UPCE (m = 1 or 66), note: When the data length is 6, the System Character (NSC) 0 inserts automatically at 6					

When the data lengths are 7,8,11, and 12, the first system character (NSC) d1 must be 0

Whether the input data length is 6,7,8,11,12 or 12

Whether the input data length is 6,7,8,11, or 12, the barcode readable character (HRI) displays only 6 bits of data, not including the system character (NSC) or verification code;

The transfer relationship between transmission data and print data is as follows:

Data transferred										Printed data					
d2	d3	d4	d5	d6	d7	d8	d9	d10	d11	d1	d2	d3	d4	d5	d6
0~9	0~9	0	0	0	-	-	0~9	0~9	0~9	d2	d3	d9	d10	d11	0
0~9	0~9	1	0	0	-	-	0~9	0~9	0~9	d2	d3	d9	d10	d11	1
0~9	0~9	2	0	0	-	-	0~9	0~9	0~9	d2	d3	d9	d10	d11	2
0~9	0~9	3~9	0	0	-	-	-	0~9	0~9	d2	d3	d4	d10	d11	3
0~9	0~9	0~9	1~9	0	-	-	-	-	0~9	d2	d3	d4	d5	d11	4
0~9	0~9	0~9	0~9	1~9	-	-	-	-	5~9	d2	d3	d4	d5	d6	d11

When d6 is 1 to 9, d7, d8, d9, d10 is 0 and d11 is 5 to 9 should be guaranteed

Start and end characters are automatically inserted

When printing EAN13 (m = 2 or 67), note:

Whether the input data length is 12 or 13, the check bit is automatically inserted or error-corrected

Start character, middle separator, and end character is automatically inserted

When printing EAN8 (m = 3 or 68), note:

Checkbits are automatically inserted or error corrected, whether the input data length is 7 or 8

Start character, middle separator, and end character is automatically inserted

When printing a CODE39 (m = 4 or 69), note:

When d1 or dn is not the start / end character "*", the encoder automatically inserts the "*"

When "*" is encountered in the middle of the data, the encoder regards it as the end character and the rest of the data as ordinary data processing;

Checkbits are not calculated and added automatically

When printing ITF25 (m = 5 or 70), note:

Start and end characters are automatically inserted

Checkbits are not calculated and added automatically

When printing a CODABAR (NW-7) (m = 6 or 71), note:

Start and end characters will not be automatically inserted and need to be added manually, by the user, either "A" ~ "D" or "a" ~ "d"

Checkbits are not calculated and added automatically

When printing a CODE93 (m = 72), note:

Start and end characters are automatically inserted

Two check codes are automatically calculated and inserted

When setting the barcode-readable characters (HRI) print, set no HRI

characters representing the start / end

When the barcode-readable characters (HRI) prints are set, the control characters are replaced with spaces

When selecting CODE128 (m = 73):

Reference Appendix A, CODE 128 for related information and character sets.

When using CODE 128, encode with the following instructions:

① must select a character set (one of CODE A, CODE B, and CODE C) before the barcode data.

② Select the character set is done by sending the character "{" combined with another character; the ASCII code character

The "{" is done by sending the character "{" twice in succession.

Special characters, to send the data

ASCII code hex code decimal code

special character	send data		
	ASCII code	hexadecimal code	decimal code
SHIFT	{S	7B,53	123, 83
CODEA	{A	7B,41	123, 65
CODEB	{B	7B,42	123, 66
CODEC	{C	7B,43	123, 67
FNC1	{1	7B,31	123, 49
FNC2	{2	7B,32	123, 50
FNC 3	{3	7B,33	123, 51
FNC 4	{4	7B,34	123, 52
"{"	{{	7B,7B	123, 123

[Instance] For example, printing " No.123456"

In this instance, the printer first prints with CODE B the " No." Then print the remaining numbers with CODE C:

GS k 73 10 123 66 78 111 46 123 67 12 34 56



CODE 128:

1b 40 1d 48 02 1d 68 64 1d 77 03

1d 6b 49 0A 7B 42 4E 6F 2E 7B 43 0C 22 38

If character set selection is not at the front of barcode data, the printer stops processing of this command and remains

The data below is processed as ordinary data.

If "{" and the character following it are not the combination specified above, then the printer stops the command

And the remaining data is processed as ordinary data.

If the printer receives characters that are not barcode character set data, the printer stops the processing of this command and remains

The data below is processed as ordinary data.

When a printer prints HRI characters, it does not print shift

	characters and character set selection data. The HRI characters for the functional characters are not printed. The HRI characters for the control characters (<00> H to <1F> H and <7F> H) are also not printed either; <Other> Must ensure that the bar code left and right gap. Gaps vary according to the barcode type.
Example of use	<pre> 1b 40 1d 48 02 1d 68 64 1d 77 03 30 0D 0A 1d 6b 00 30 31 32 33 34 35 36 37 38 39 31 00 31 0D 0A 1d 6b 01 30 31 32 33 34 35 36 37 38 39 31 00 32 0D0A 1d 6b 02 30 31 32 33 34 35 36 37 38 39 31 32 00 33 0D 0A 1d 6b 03 30 31 32 33 34 35 36 37 00 34 0D 0A 1D 6B 04 30 31 32 41 42 20 24 25 2B 2D 2E 2F 00 35 0D 0A 1d 6b 05 30 31 32 33 34 35 36 37 38 39 31 32 00 36 0D 0A 1d 6b 06 2D 31 32 42 24 2B 2D 2E 00 1d 6b 06 43 31 32 33 34 35 36 34 38 39 00 36 35 0D 0A 1d 6b 41 0c 31 32 33 34 35 36 37 38 39 30 31 32 36 36 0D 0A 1d 6b 42 0c 30 32 33 34 35 36 30 30 30 30 38 39 36 37 0D 0A 1d 6b 43 0c 30 32 33 34 35 36 30 30 30 30 38 39 36 38 0D 0A 1d 6b 44 08 30 32 33 34 35 36 30 30 36 39 20 20 4e 4f 20 24 25 2b 2d 2e 2f 31 32 33 34 35 36 30 30 0D 0A 1d 6b 45 11 4e 4f 20 24 25 2b 2d 2e 2f 31 32 33 34 35 36 30 30 37 30 20 20 20 30 32 33 34 35 36 30 30 C5 BC CA FD 0D 0A 1d 6b 46 09 30 31 32 33 34 35 36 30 30 37 31 0d 0a 1d 6b 47 05 32 33 34 35 36 37 32 0d 0a 1d 6b 48 0b 32 33 34 35 36 41 42 2e 2f 2b 2c 37 33 0d0a 1d 6b 49 0A 7B 42 4E 6F 2E 7B 43 0C 22 38 Code 128 : 1b 40 1d 48 02 1d 68 64 1d 77 03 37 33 0d0a 1d 6b 49 0A 7B 42 4E 6F 2E 7B 43 0C 22 38 </pre>

6.9QR code print instruction

6.9.1Set the module type of QR code

Command name	Sets the module type of the QR code
Instruction code	ASCII : GS (k pL pH cn fn n Decimal : 29 40 107 pL pH cn fn n Hex : 1D 28 6b pL pH cn fn n
Function description	Sets the module type of the QR code
Parameter range	pL=3, pH=0 cn=49 fn=67 $0 \leq n \leq 16$
Defaults	n=3
Support model	All model
Precautions	
Example of use	None

6.9.2Set the error correction level error of the QR code

Command name	Sets the error correction level error for the QR code		
Instruction code	ASCII : GS (k pL pH cn fn n Decimal : 29 40 107 pL pH cn fn n Hex : 1D 28 6b pL pH cn fn n		
Function description	Sets the error correction level error for the QR code		
Parameter range	pL=3, pH=0 cn=49 fn=69 $48 \leq n \leq 51$		
Defaults	n=48		
Support model	All model		
Precautions	Sets the error correction level error for the QR code		
	n	function	consult: The recovery of the case presumably represents (%)
	48	Error correction level	7

		error L	
	49	Error correction level error m	15
	50	Error correction level error q	25
	51	Error correction level error h	30
Example of use	None		

6.9.3 Print the QR code

Command name	Print the QR code
Instruction code	ASCII : GS (k pL pH cn fn m Decimal : 29 40 107 pL pH cn fn m Hex : 1D 28 6b pL pH cn fn m
Function description	Print the QR code
Parameter range	pL=3, pH=0 cn=49 fn=81 m=48
Defaults	None
Support model	All model
Precautions	Print the QR code。
Example of use	1b 40 1d 28 6b 03 00 31 43 03 1d 28 6b 03 00 31 45 30 1d 28 6b 06 00 31 50 30 41 42 43 1b 61 01 1d 28 6b 03 00 31 52 30 1d 28 6b 03 00 31 51 30

6.9.4 Print a QR code

Command name	Print a QR code
Instruction code	ASCII : GS k m v r nL nH d1...dk Decimal : 29 107 97 v r nL nH d1...dk Hex : 1D 6B 61 v r nL nH d1...dk
Function	Print a QR code

description	v means the specification of the QR code, v=0 means the specification of the QR code is automatically selected r stands for error correction level nL nH indicates the data length d1...dk indicates the QR code data to be printed
Parameter range	$0 \leq v \leq 17$ $1 \leq r \leq 4$ $k = nL + 256 * nH$
Defaults	None
Support model	All model
Precautions	Print the QR code.
Example of use	1b 40 1B 61 01 1D 6B 61 08 04 18 00 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0D 0A

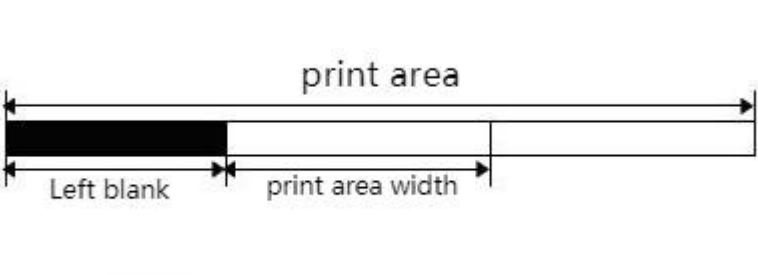
6.10print setting instructions

6.10.1Set line spacing to n points

Command name	Set line spacing to n points
Instruction code	ASCII : ESC 3 n Decimal : 27 51 n Hex : 1B 33 n
Function description	Set the row spacing to be n points
Parameter range	$0 \leq n \leq 255$
Defaults	n = 33
Support model	All models
Precautions	The line spacing is indicated as follows:  If the row spacing is less than the maximum character height in a row, the row spacing is equal to the maximum character height If ESC 2, ESC @, printer reset, printer power, row spacing restored to default value
Example of use	1b 40 1b 33 30 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a

	CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a 1b 33 50 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a
--	--

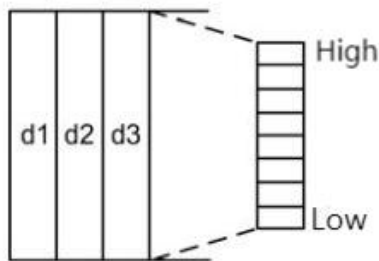
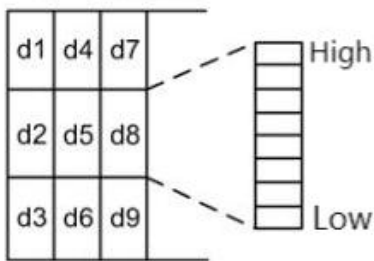
6.10.2Set the left margin

Command name	Set the print location
Instruction code	ASCII : GS L nL nH Decimal: 2,976 n L n H Hex: 1D 4C nL nH
Function description	Set the left blank quantity as the (nL + nH 256) point
Parameter range	$0 \leq nL \leq 255, 0 \leq nH \leq 255$
Defaults	None
Support model	All models
Precautions	The command is only valid when processed at the beginning of a row. The legend is shown as follows:  If the setting is beyond the printable range, then the maximum printable unit value is used
Example of use	1b 40 1d 4c 48 00 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a

6.11Graphic print instructions

6.11.1Graphic vertical modulo data filling

Command name	The graph is filled with data vertically
--------------	--

Instruction code	ASCII : ESC * m Hl Hh [d]k Decimal : 27 42 m Hl Hh [d]k Hex : 1B 2A m Hl Hh [d]k																				
Function description	Print longitudinal mold image data with parameters as follows: The m is in the dot plot format: <table><tr><td>M</td><td></td><td>horizontal scale</td><td>vertical scale</td></tr><tr><td>0 8</td><td>Point single density</td><td>× 2</td><td>× 3</td></tr><tr><td>1 8</td><td>Point double density</td><td>× 1</td><td>× 3</td></tr><tr><td>32 24</td><td>Point single density</td><td>× 2</td><td>× 1</td></tr><tr><td>33 24</td><td>Point double density</td><td>× 1</td><td>× 1</td></tr></table> Hl and Hh are the number of horizontal points (Hl + 256 Hh) [d] k is the dot plot data The k is used to indicate the number of dot plot data bytes and does not participate in transmission	M		horizontal scale	vertical scale	0 8	Point single density	× 2	× 3	1 8	Point double density	× 1	× 3	32 24	Point single density	× 2	× 1	33 24	Point double density	× 1	× 1
M		horizontal scale	vertical scale																		
0 8	Point single density	× 2	× 3																		
1 8	Point double density	× 1	× 3																		
32 24	Point single density	× 2	× 1																		
33 24	Point double density	× 1	× 1																		
Parameter range	XX58: m = 0、 1、 32、 33 $1 \leq Hl + Hh \times 256 \leq 384$ $0 \leq d \leq 255$ At k = Hl + Hh 256 (when m = 0,1) 1 = k = (Hl + Hh 256) 3 (when m = 32,33) XX80: m = 0、 1、 32、 33 $1 \leq Hl + Hh \times 256 \leq 576$ $0 \leq d \leq 255$ At k = Hl + Hh 256 (when m = 0,1) 1 = k = (Hl + Hh 256) 3 (when m = 32,33)																				
Defaults	None																				
Support model	All model																				
Precautions	The corresponding bit of [d] k means the point printing and the corresponding bit of 0 means the point does not print The portion of the horizontal image beyond the print area will be ignored The relationship between the dot plot data and the printing effect is as follows: 8 points mode  Dot plot data(bitmap) 24 points mode  Dot plot data(bitmap) This instruction only fills the print cache. The printing of the image starts after receiving the print instruction, and the print cache is emptied after the image is printed If the image height required to print is large, it can be divided into several images with 8 (m = 0,1) or 24 (m = 32,33) points and printed respectively																				

	After filling the graphic data, you can continue to fill other information so that the drawing is printed with other information After filling the point map, ESC J (n = 24) instruction is generally used to print, or LF instruction to print, but LF instruction will trigger the paper feeding operation (according to the row spacing), making the multiple line image intermittent, can set the row spacing of 0, there will not be too much paper.(The pin printer will offset, if a line occurs in the middle, send data continuously)
Example of use	<pre>1B 40 1b 2a 00 0C 00 FF FF FF FF FF FF FF FF FF FF FF 1B 33 00 0A</pre>

6.11.2Picture horizontal modulo data printing

Command name	The picture is printed horizontally to mold data																				
Instruction code	ASCII : GS v 0 Decimal : 29 118 48 m xL xH yL yH [d]k Hex : 1D 76 30 m xL xH yL yH [d]k																				
Function description	Print horizontal mold image data with parameters as follows: The way of m is a bitmap: The m-mode horizontal scale is a vertical scale <table><tr><th>m</th><th>Mode</th><th>Horizontal proportions</th><th>Vertical scale</th></tr><tr><td>0,48</td><td>Normal</td><td>×1</td><td>×1</td></tr><tr><td>1,49</td><td>X Width</td><td>×2</td><td>×1</td></tr><tr><td>2,50</td><td>X High</td><td>×1</td><td>×2</td></tr><tr><td>3,51</td><td>X width& X High</td><td>×2</td><td>×2</td></tr></table> For xL, xH, the horizontal direction is the number of bytes (xL + xH 256) The yL and yH are the vertically oriented points (yL + yH 256) [d] k is the dot plot data The k is the number of bytes of the dot plot data. The k is used for signal without transmission	m	Mode	Horizontal proportions	Vertical scale	0,48	Normal	×1	×1	1,49	X Width	×2	×1	2,50	X High	×1	×2	3,51	X width& X High	×2	×2
m	Mode	Horizontal proportions	Vertical scale																		
0,48	Normal	×1	×1																		
1,49	X Width	×2	×1																		
2,50	X High	×1	×2																		
3,51	X width& X High	×2	×2																		
Parameter range	XX58: $0 \leq m \leq 3; 48 \leq m \leq 51$ $1 \leq xL + xH \times 256 \leq 48$ $0 \leq yL \leq 255, 0 \leq yH \leq 255$ $0 \leq d \leq 255$ $k = (Hl + Hh \times 256) \times (yL + yH \times 256)$ XX80: $0 \leq m \leq 3; 48 \leq m \leq 51$ $1 \leq xL + xH \times 256 \leq 72$ $0 \leq yL \leq 255, 0 \leq yH \leq 255$ $0 \leq d \leq 255$ $k = (Hl + Hh \times 256) \times (yL + yH \times 256)$																				
Defaults	None																				
Support	All models																				

model																	
Precautions	The corresponding bit of [d] k means the point printing and the corresponding bit of 0 means the point does not print If the image level bytes exceed the print area, the excess will be ignored This instruction is executed to the image size, not affected by the row spacing of ESC 2 and ESC 3 After this instruction is executed, the print coordinates are reset to the left distance position, and the image content is emptied The relationship between the bitmap data and the printing effect is as follows: <table><tr><td>d1</td><td>d2</td><td>.....</td><td>dx</td></tr><tr><td>d(x+1)</td><td>d(x+2)</td><td>.....</td><td>d(x×2)</td></tr><tr><td> </td><td> </td><td>.....</td><td> </td></tr><tr><td>.....</td><td>d(k-2)</td><td>d(k-1)</td><td>dk</td></tr></table> MSB LSB MSB LSB MSB LSB MSB LSB This instruction has a printing function, while transmitting data while printing, no need to use the printing instruction	d1	d2		dx	d(x+1)	d(x+2)		d(x×2)						d(k-2)	d(k-1)	dk
d1	d2		dx														
d(x+1)	d(x+2)		d(x×2)														
																	
.....	d(k-2)	d(k-1)	dk														
Example of use	1B 40 1d 76 30 00 03 00 09 00 FF																

7. Printer Status and Settings

7.1 Out of paper status

Command name	Check the absence of paper
Instruction code	Decimal: 16 04 01 Hex: 10 04 01
Function description	Check the absence of paper
Parameter range	
Defaults	
Support model	
Precautions	When the printer is out of paper, it automatically returns the missing paper state "EF 23 1A" once a second. When the paper is successful, the missing paper state "FE 23 12" returns only once.
Example of use	Send check missing paper instruction: 10 04 01, send data once and return it once Return data: FE 23 12 (printer with paper)Returns dataReturns data

	Returns data: FE 23 12 (Printer has paper) EF 23 1A (the printer is out of paper)
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7.2 print status

Command name	Print status
Instruction code	
Function description	Print status
Parameter range	
Defaults	
Support model	
Precautions	Send printer Data printer printing completed at 500ms (milliseconds) no data sending printer automatically returns data: FC 4F 4B (printing completed) Paper paper during sending data printing returns FC 6E 6F (print failure)
Example of use	

7.3 Set the printer serial port baud rate

Command name	Set the printer Baud rate			
Instruction code	ASCII : US - U 1 m Decimal: 3,1,45,8,51 m Hex: 1F 2D 55 01 m			
Function description	M< 29			
	Pot rate corresponding value			
	M price	Baud rate	M price	Baud rate
	0	1200	15	307200
	1	2400	16	460800
	2	3600	17	614400
	3	4800	18	921600
	4	7200	19	1228800
	5	9600	20	1843200
	6	14400		
	7	19200		
	8	28800		
	9	38400		
	10	57600		

	11	76800		
	12	115200		
	13	153600		
	14	230400		
Parameter range				
Defaults	M=5 A default baud rate of 9600			
Support model				
Precautions	Order issued: 1F 2D 55 01 m Return data: 1F 2D 55 01 m (return data format and distribution the same) Return data: 1F 2D 55 01 m (the return data format is the same as the issue)			
Example of use				

7.4 Set the default serial port open or closed state when power off

Command name	Set the serial port communication on and off
Instruction code	ASCII : US - q 1 m Decimal: 31 45 113 1 m Hex: 1F 2D 71 01 m
Function description	M=0, open the serial port, M= 1, and close the serial port In the process of serial communication between the device and the device, if the serial port closing mode is set, the printer will not process any data, to prevent data interference without printing, affecting the normal operation of the printer.
Parameter range	
Defaults	M=0 The default serial port is open
Support model	
Precautions	Power-on again is effective. This instruction is just used in the early setting, and it does not need to be called during the printing process.
Example of use	Issue instruction: 1F 2D 71 01 m The printer prints out the "Successfully Set The Uart open state is Close" after issuing the setup data Return data: 1F 2D 71 01 m (return data format and distribution the same)

7.5 Set the serial port status (this command will not be saved when power off)

Command name	
Instruction code	ASCII : US w m Decimal: 31,119 m Hex: 1F 77 m
Function description	M=0, open the serial port, M= 1, and close the serial port This instruction is used during the printing process, before sending the serial port set to open mode, then send the print data, and then close the serial port.
Parameter range	
Defaults	
Support model	
Precautions	Open the serial port first- - -send the print data- -and then close the serial port
Example of use	Issue instruction: 1F 77 00 (open the serial port first) 1b 40 1b 33 30 CF C3 C3 C5 B4 EF C6 D5 B5 E7 D7 D3 BF C6 BC BC D3 D0 CF DE B9 AB CB BE 0d 0a 1F 77 01 (close the serial port again)

7.6 Set whether to feed paper, the number of lines to feed paper, and how long to feed paper after the end of data

Command name	Set whether to feed paper, feed paper function, end how long the data to feed paper
Instruction code	ASCII : US - 5 n m k tL tH Decimal: 31 45 53 4 m k tL tH Hex: 1F 2D 35 04 m k tL tH
Function description	After the data is printed, set whether to enter the paper, the number of paper lines, and how long the data starts to enter the paper
Parameter range	10<=tL + tH *256 <=1000;n=0,1;1<=k<=256; In m: 0, no paper, 1, no paper k: Number of paper lines

	The tL + tH * 256: judge the time of data, default 200ms
Defaults	m=0,k=2,tL+tH*256 = 200=C8 00
Support model	
Precautions	Mainly used for small ticket mode
Example of use	Issue command: 1F 2D 35 04 00 05 C8 00 Return data: 1F 2D 35 04 00 05 C8 00 (the return data format is the same as the one sent)

7.7Full cut paper

Command name	Full cut paper
Instruction code	Decimal : 27 105 Hex : 1B 69
Function description	The cutter cuts the paper completely
Parameter range	None
Defaults	None
Precautions	None
Example of use	1B 40 30 30 30 0D 0A 1B 69

7.8half cut paper

Command name	Full cut paper
Instruction code	Decimal : 27 109 Hex : 1B 6D
Function description	The cutter is half-cut, and there is still a little bit of paper left, and the paper is gently pulled out by hand.
Parameter range	None
Defaults	None
Precautions	None
Example of use	1B 40 30 30 30 0D 0A 1B 6D

7.9 Set whether to automatically cut the knife

Command name	Set whether to automatically cut the knife
Instruction code	ASCII : US - a 2 m n Decimal : 31 45 97 2 m n Hex : 1F 2D 61 02 m n
Function description	If the automatic cutter is set to be turned on, the paper will be automatically cut without adding the cutter command after the printing is completed. If you set the door closing automatic cutter, you need to add a cutter command to cut the paper after the printing is completed. Mainly used for driver printing, some printer drivers do not have a cutter function for printing. Need to add automatic cutter action
Parameter range	M = 0 (turn off automatic cutter) M = 1 (turn on automatic cutter) N = 0 (half cut) N = 1 (full cut)
Defaults	
Support model	
Precautions	Issue command: 1F 2D 61 02 01 00 (the threshold is 0x0150) Return data: 1F 2D 61 02 01 00 (the return data format is the same as the one sent)
Example of use	

7.10 Set print mode

Command name	Set print mode
Instruction code	ASCII : US - M l m Decimal : 31 45 77 1 m Hex : 1F 2D 4D 01 m
Function description	Set print mode
Parameter range	m=0x01 (label mode), m=0x02 (small ticket mode)
Defaults	m=0x01 (label mode)
Support model	
Precautions	Issue command: 1F 2D 4D 01 01 Return data: 1F 2D 4D 01 01 (the return data format is the same as the one sent)
Example of use	

8. Comprehensive printing example

8.1 barcode printing

1B 40 1B 61 01 1D 48 02 1D 68 50 00 1D 77 02 00 1D 6B 49 0b 31 32 33 34 35 36 37 38 39 31 30

1B 40 Initialize the printer

1B 61 00 // barcode is to the left

1 The B 6101 // Barcode is in the center

1B 61 02 // barcode is to the left

The 1D 48 01 // data are displayed above the barcode

The 1 The D 4802 // data are displayed below the barcode

1D 48 03 // data are displayed up and below the barcode

1D 48 00 // does not display numbers only the barcode

1D 68 50 00 // 1D 68 Set the barcode height from 50 00 to 80 height, with the height range from 10-200

1D 77 02 00 // 1D 77 Set the barcode width 03 00 to 2 width, width range in 1-6

1D 6B 49 // Barcode type CODE128

0B 31 32 33 34 35 36 37 38 39 31 30 // 0B is the data length 11, bar code data 31323333435363738393130 content "12345678910"

Note that the bar code does not support Chinese characters and Chinese characters



8.2 Text printing

1B 40 1B 33 10 1D 21 11 1B 61 01 BB B6 D3 AD B9 E2 C1 D9 0D 0A

1B 40 1B 33 10 // Set line high distance 10 line range 10,20,30,40,50,60
1B 40 1B 33 20
1B 40 1B 33 30
1B 40 1B 33 40
1B 40 1B 33 50
1B 40 1B 33 60

1D 21 00 // Normal font size
1D 21 11 // font size doubled
1D 21 10 // font width
1D 21 01 // font height

1B 61 00 // Text left alignment
1B 61 01 // Text text alignment
1B 61 02 // Text Right alignment

BB B6 D3 AD B9 E2 C1 D9 Text Print Content "Welcome"

The 0D 0A End character, which can also be used as a line change

8.3 QR code printing

1b 40 // Fixed
1d 28 6b 03 00 31 43 03 // QR code size 43 02,43 03,43 04,43 05,43 06,43 07,43 08
1d 28 6b 03 00 31 45 30 // Fixed
1d 28 6b 06 00 31 50 30 41 42 43 // 06 00 Data Length (31 50 30 41 42 43) 6 Data Length,
31 50 30 Fixed, 41 42 43 qr code content "ABC"
1b 61 01 // 00 QR code on the left 01 QR code in the middle 10 QR code on the right
1d 28 6b 03 00 31 52 30 // Fixed
1d 28 6b 03 00 31 51 30 // Fixed

Text content printing
1b 40 // Fixed
1b 61 01 // 00 House left 01 house center 10 house right
1d 21 00 // 00 Normal 01 x width 10 x height 11 x width height
C9 A8 D2 BB C9 A8 B9 D8 D7 A2 // Text content "Scan attention"
0d 0a // Line change

1b 40
1d 28 6b 03 00 31 43 08
1d 28 6b 03 00 31 45 30
1d 28 6b 06 00 31 50 30 41 42 43
1b 61 01
1d 28 6b 03 00 31 52 30
1d 28 6b 03 00 31 51 30
1b 40 1d 21 00

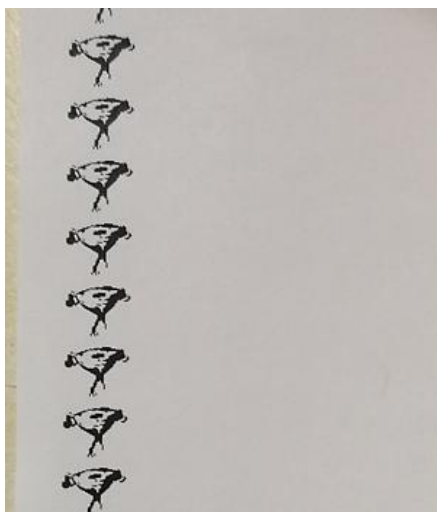
C9 A8 D2 BB C9 A8 B9 D8 D7 A2 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 1b 69



8.4 Transverse picture and picture printing

[illegible]

Here and behind it is the picture data.



8.6 Print label QR code

Command example:

1A 5B 01 00 00 00 00 80 01 EA 00 00

1A 31 00 05 04 15 00 15 00 04 00 53 54 4A 41 31 30 33 31 39 31 31 30 30 30 30 31 00

1A 54 01 00 00 b0 00 00 60 00 11 53 54 4A 41 31 30 33 31 39 31 31 30 30 30 30 31 00

1A 5D 00 1A 4F 00 1b 6d

Instruction parsing

1A 5B 01 00 00 00 00 80 01 EA 00 00 //Set the printing range "00 00 00 00" sets the initial printing position of the X and Y axes, the default is 0, and "80 01 EA 00" sets the width and height of the printing area. 00 end

1A 31 00 05 02 15 00 15 00 04 00 53 54 4A 41 31 30 33 31 39 31 31 30 30 30 30 31 00 //Print a QR code

1A 31 00 Print a QR code label

05 Set the value range of the QR code version [0-20], the larger the value, the denser the QR code.

03 Error correction level value range [1-4] Low error correction, more data, high error correction, less data

15 00 15 00 X, Y axis printing position

04 QR code size value range [1-7]

00 Rotation angle 【0-3】 0, 90°, 180°, 270°

53 54 4A 41 31 30 33 31 39 31 31 30 30 30 30 31 QR code content "STJA103191100001"

00 Cut off data flow

1A 54 01 00 00 b0 00 00 60 00 11 53 54 4A 41 31 30 33 31 39 31 31 30 30 30 30 31 00 Print Text Content Label Instruction

1A 54 01 Text printing label start

00 00 b0 00X, Y axis printing position

00 60 00 Fixed

11, print text size, 11, 22, 33, 44, 55, 66 have 6 font sizes, all of which are multiplied

1A 5D 00 //End of label printing

1A 4F 00 //Print the content onto paper.

1b 6d cut paper

